

Thermodynamics

Instructor

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Chapter 3 – Problems

**Problem numbers correspond to
5th Edition**

3-26

Complete this table for H₂O:

$T, ^\circ\text{C}$	P, kPa	$\nu, \text{m}^3/\text{kg}$	Phase description
50		4.16	
	200		Saturated vapor
250	400		
110	600		

$T, ^\circ\text{C}$	P, kPa	$\nu, \text{m}^3/\text{kg}$	Phase description
50	12.352	4.16	<i>Saturated mixture</i>
120.21	200	0.8858	Saturated vapor
250	400	0.5952	<i>Superheated vapor</i>
110	600	0.001051	<i>Compressed liquid</i>

3–30 Complete this table for H₂O:

$T, ^\circ\text{C}$	P, kPa	$h, \text{kJ/kg}$	x	Phase description
	200		0.7	
140		1800		
	950		0.0	
80	500			
	800	3162.2		

T, °C	P, kPa	h, kJ / kg	x	Phase description
<i>120.21</i>	200	<i>2045.8</i>	0.7	<i>Saturated mixture</i>
140	<i>361.53</i>	1800	<i>0.565</i>	<i>Saturated mixture</i>
<i>177.66</i>	950	<i>752.74</i>	0.0	<i>Saturated liquid</i>
80	500	<i>335.37</i>	- - -	<i>Compressed liquid</i>
<i>350.0</i>	800	3162.2	- - -	<i>Superheated vapor</i>

3-31 Complete this table for refrigerant-134a:

$T, ^\circ\text{C}$	P, kPa	$\nu, \text{m}^3/\text{kg}$	Phase description
-8	320		
30		0.015	
	180		Saturated vapor
80	600		

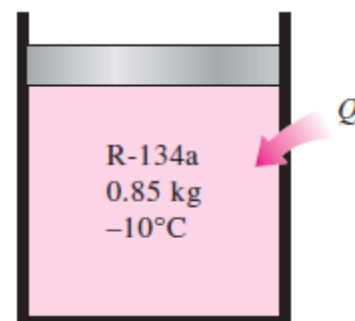
T, °C	P, kPa	$\nu, \text{m}^3 / \text{kg}$	Phase description
-8	320	<i>0.0007569</i>	<i>Compressed liquid</i>
30	<i>770.64</i>	0.015	<i>Saturated mixture</i>
<i>-12.73</i>	180	<i>0.11041</i>	Saturated vapor
80	600	<i>0.044710</i>	<i>Superheated vapor</i>

3-32 Complete this table for refrigerant-134a:

$T, ^\circ\text{C}$	P, kPa	$u, \text{kJ/kg}$	Phase description
20		95	
-12			Saturated liquid
	400	300	
8	600		

T, °C	P, kPa	u, kJ / kg	Phase description
20	572.07	95	<i>Saturated mixture</i>
-12	185.37	35.78	Saturated liquid
86.24	400	300	<i>Superheated vapor</i>
8	600	62.26	<i>Compressed liquid</i>

3-37 A piston–cylinder device contains 0.85 kg of refrigerant-134a at -10°C . The piston that is free to move has a mass of 12 kg and a diameter of 25 cm. The local atmospheric pressure is 88 kPa. Now, heat is transferred to refrigerant-134a until the temperature is 15°C . Determine (a) the final pressure, (b) the change in the volume of the cylinder, and (c) the change in the enthalpy of the refrigerant-134a.



Analysis (a) The final pressure is equal to the initial pressure, which is determined from

$$P_2 = P_1 = P_{\text{atm}} + \frac{m_p g}{\pi D^2/4} = 88 \text{ kPa} + \frac{(12 \text{ kg})(9.81 \text{ m/s}^2)}{\pi(0.25 \text{ m})^2/4} \left(\frac{1 \text{ kN}}{1000 \text{ kg}\cdot\text{m/s}^2} \right) = \mathbf{90.4 \text{ kPa}}$$

(b) The specific volume and enthalpy of R-134a at the initial state of 90.4 kPa and -10°C and at the final state of 90.4 kPa and 15°C are (from EES)

$$\nu_1 = 0.2302 \text{ m}^3/\text{kg} \quad h_1 = 247.76 \text{ kJ/kg}$$

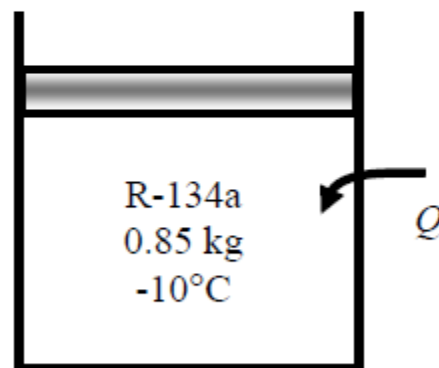
$$\nu_2 = 0.2544 \text{ m}^3/\text{kg} \quad h_2 = 268.16 \text{ kJ/kg}$$

The initial and the final volumes and the volume change are

$$\mathcal{V}_1 = m \nu_1 = (0.85 \text{ kg})(0.2302 \text{ m}^3/\text{kg}) = 0.1957 \text{ m}^3$$

$$\mathcal{V}_2 = m \nu_2 = (0.85 \text{ kg})(0.2544 \text{ m}^3/\text{kg}) = 0.2162 \text{ m}^3$$

$$\Delta \mathcal{V} = 0.2162 - 0.1957 = \mathbf{0.0205 \text{ m}^3}$$



(c) The total enthalpy change is determined from

$$\Delta H = m(h_2 - h_1) = (0.85 \text{ kg})(268.16 - 247.76) \text{ kJ/kg} = \mathbf{17.4 \text{ kJ/kg}}$$

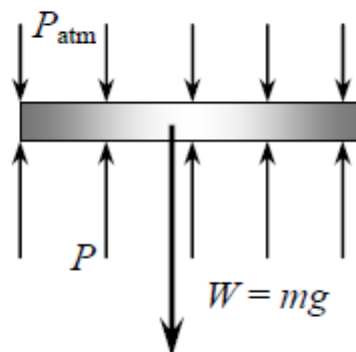
3–50 Water is being heated in a vertical piston–cylinder device. The piston has a mass of 20 kg and a cross-sectional area of 100 cm². If the local atmospheric pressure is 100 kPa, determine the temperature at which the water starts boiling.

Analysis The pressure in the cylinder is determined from a force balance on the piston,

$$PA = P_{\text{atm}}A + W$$

or,

$$\begin{aligned} P &= P_{\text{atm}} + \frac{mg}{A} \\ &= (100 \text{ kPa}) + \frac{(20 \text{ kg})(9.81 \text{ m/s}^2)}{0.01 \text{ m}^2} \left(\frac{1 \text{ kPa}}{1000 \text{ kg/m} \cdot \text{s}^2} \right) \\ &= 119.61 \text{ kPa} \end{aligned}$$



The boiling temperature is the saturation temperature corresponding to this pressure,

$$T = T_{\text{sat}@119.61 \text{ kPa}} = \mathbf{104.7^\circ\text{C}} \quad (\text{Table A-5})$$

3-51 A rigid tank with a volume of 2.5 m^3 contains 15 kg of saturated liquid–vapor mixture of water at 75°C . Now the water is slowly heated. Determine the temperature at which the liquid in the tank is completely vaporized. Also, show the process on a T - ν diagram with respect to saturation lines.

Answer: 187.0°C

Analysis This is a constant volume process ($\nu = V/m = \text{constant}$), and the specific volume is determined to be

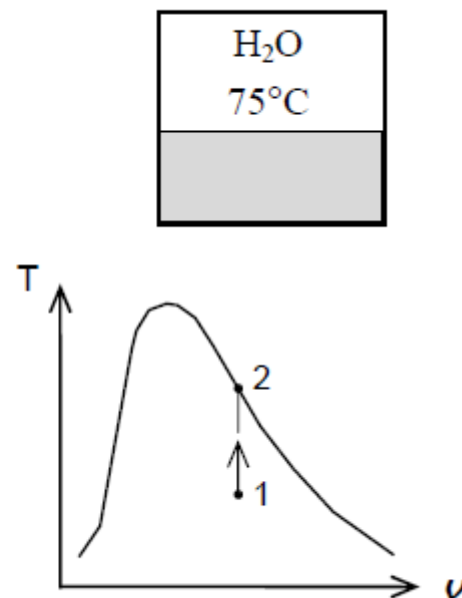
$$\nu = \frac{V}{m} = \frac{2.5 \text{ m}^3}{15 \text{ kg}} = 0.1667 \text{ m}^3/\text{kg}$$

When the liquid is completely vaporized the tank will contain saturated vapor only. Thus,

$$\nu_2 = \nu_g = 0.1667 \text{ m}^3/\text{kg}$$

The temperature at this point is the temperature that corresponds to this ν_g value,

$$T = T_{\text{sat}@ \nu_g = 0.1667 \text{ m}^3/\text{kg}} = \mathbf{187.0^\circ\text{C}} \quad (\text{Table A-4})$$



3-52 A rigid vessel contains 2 kg of refrigerant-134a at 800 kPa and 120°C. Determine the volume of the vessel and the total internal energy. *Answers: 0.0753 m³, 655.7 kJ*

Properties The properties of R-134a at the given state are (Table A-13).

$$\left. \begin{array}{l} P = 800 \text{ kPa} \\ T = 120^\circ \text{C} \end{array} \right\} \begin{array}{l} u = 327.87 \text{ kJ/kg} \\ \nu = 0.037625 \text{ m}^3/\text{kg} \end{array}$$

Analysis The total volume and internal energy are determined from

$$V = m\nu = (2 \text{ kg})(0.037625 \text{ m}^3/\text{kg}) = \mathbf{0.0753 \text{ m}^3}$$

$$U = mu = (2 \text{ kg})(327.87 \text{ kJ/kg}) = \mathbf{655.7 \text{ kJ}}$$

R-134a
2 kg
800 kPa
120°C

3-54 A 0.5-m³ vessel contains 10 kg of refrigerant-134a at -20°C. Determine (a) the pressure, (b) the total internal energy, and (c) the volume occupied by the liquid phase.

Answers: (a) 132.82 kPa, (b) 904.2 kJ, (c) 0.00489 m³

Analysis (a) The specific volume of the refrigerant is

$$\nu = \frac{V}{m} = \frac{0.5 \text{ m}^3}{10 \text{ kg}} = 0.05 \text{ m}^3/\text{kg}$$

At -20°C, $\nu_f = 0.0007362 \text{ m}^3/\text{kg}$ and $\nu_g = 0.14729 \text{ m}^3/\text{kg}$ (Table A-11). Thus the tank contains saturated liquid-vapor mixture since $\nu_f < \nu < \nu_g$, and the pressure must be the saturation pressure at the specified temperature,

$$P = P_{\text{sat}@-20^\circ\text{C}} = \mathbf{132.82 \text{ kPa}}$$

(b) The quality of the refrigerant-134a and its total internal energy are determined from

$$x = \frac{\nu - \nu_f}{\nu_{fg}} = \frac{0.05 - 0.0007362}{0.14729 - 0.0007362} = 0.3361$$

$$u = u_f + xu_{fg} = 25.39 + 0.3361 \times 193.45 = 90.42 \text{ kJ/kg}$$

$$U = mu = (10 \text{ kg})(90.42 \text{ kJ/kg}) = \mathbf{904.2 \text{ kJ}}$$

(c) The mass of the liquid phase and its volume are determined from

$$m_f = (1 - x)m_t = (1 - 0.3361) \times 10 = 6.639 \text{ kg}$$

$$V_f = m_f \nu_f = (6.639 \text{ kg})(0.0007362 \text{ m}^3/\text{kg}) = \mathbf{0.00489 \text{ m}^3}$$

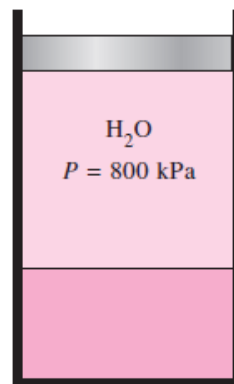
R-134a
10 kg
-20°C

3-55



A piston–cylinder device contains 0.1 m^3 of liquid water and 0.9 m^3 of water vapor in equilibrium at 800 kPa . Heat is transferred at constant pressure until the temperature reaches 350°C .

- What is the initial temperature of the water?
- Determine the total mass of the water.
- Calculate the final volume.
- Show the process on a P - v diagram with respect to saturation lines.



Analysis (a) Initially two phases coexist in equilibrium, thus we have a saturated liquid-vapor mixture. Then the temperature in the tank must be the saturation temperature at the specified pressure,

$$T = T_{\text{sat}@800 \text{ kPa}} = \mathbf{170.41^\circ\text{C}}$$

- The total mass in this case can easily be determined by adding the mass of each phase,

$$m_f = \frac{V_f}{v_f} = \frac{0.1 \text{ m}^3}{0.001115 \text{ m}^3/\text{kg}} = 89.704 \text{ kg}$$

$$m_g = \frac{V_g}{v_g} = \frac{0.9 \text{ m}^3}{0.24035 \text{ m}^3/\text{kg}} = 3.745 \text{ kg}$$

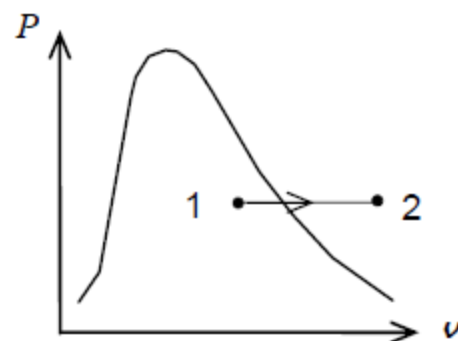
$$m_t = m_f + m_g = 89.704 + 3.745 = \mathbf{93.45 \text{ kg}}$$

- At the final state water is superheated vapor, and its specific volume is

$$\left. \begin{array}{l} P_2 = 800 \text{ kPa} \\ T_2 = 350^\circ\text{C} \end{array} \right\} v_2 = 0.35442 \text{ m}^3/\text{kg} \quad (\text{Table A-6})$$

Then,

$$V_2 = m_t v_2 = (93.45 \text{ kg})(0.35442 \text{ m}^3/\text{kg}) = \mathbf{33.12 \text{ m}^3}$$



3–60 A 0.3-m^3 rigid vessel initially contains saturated liquid–vapor mixture of water at 150°C . The water is now heated until it reaches the critical state. Determine the mass of the liquid water and the volume occupied by the liquid at the initial state. *Answers: 96.10 kg, 0.105 m^3*

Analysis This is a constant volume process ($\nu = V/m = \text{constant}$) to the critical state, and thus the initial specific volume will be equal to the final specific volume, which is equal to the critical specific volume of water,

$$\nu_1 = \nu_2 = \nu_{cr} = 0.003106\text{ m}^3/\text{kg} \quad (\text{last row of Table A-4})$$

The total mass is

$$m = \frac{V}{\nu} = \frac{0.3\text{ m}^3}{0.003106\text{ m}^3/\text{kg}} = 96.60\text{ kg}$$

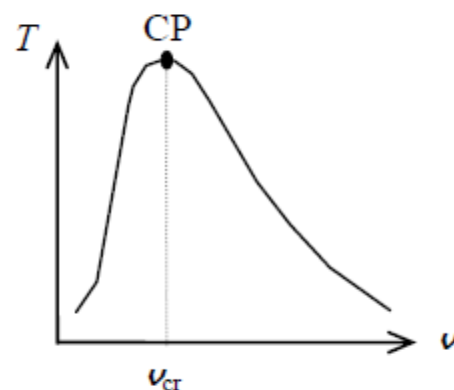
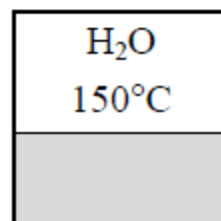
At 150°C , $\nu_f = 0.001091\text{ m}^3/\text{kg}$ and $\nu_g = 0.39248\text{ m}^3/\text{kg}$ (Table A-4). Then the quality of water at the initial state is

$$x_1 = \frac{\nu_1 - \nu_f}{\nu_{fg}} = \frac{0.003106 - 0.001091}{0.39248 - 0.001091} = 0.005149$$

Then the mass of the liquid phase and its volume at the initial state are determined from

$$m_f = (1 - x_1)m_t = (1 - 0.005149)(96.60) = \mathbf{96.10\text{ kg}}$$

$$V_f = m_f \nu_f = (96.10\text{ kg})(0.001091\text{ m}^3/\text{kg}) = \mathbf{0.105\text{ m}^3}$$



3-61 Determine the specific volume, internal energy, and enthalpy of compressed liquid water at 100°C and 15 MPa using the saturated liquid approximation. Compare these values to the ones obtained from the compressed liquid tables.

Analysis Compressed liquid can be approximated as saturated liquid at the given temperature. Then from Table A-4,

$$\begin{aligned}T = 100^{\circ}\text{C} \Rightarrow \quad \nu &\cong \nu_f @ 100^{\circ}\text{C} = 0.001043 \text{ m}^3/\text{kg} \quad (0.72\% \text{ error}) \\u &\cong u_f @ 100^{\circ}\text{C} = 419.06 \text{ kJ/kg} \quad (1.02\% \text{ error}) \\h &\cong h_f @ 100^{\circ}\text{C} = 419.17 \text{ kJ/kg} \quad (2.61\% \text{ error})\end{aligned}$$

From compressed liquid table (Table A-7),

$$\left. \begin{array}{l}P = 15 \text{ MPa} \\T = 100^{\circ}\text{C}\end{array} \right\} \begin{array}{l}\nu = 0.001036 \text{ m}^3/\text{kg} \\u = 414.85 \text{ kJ/kg} \\h = 430.39 \text{ kJ/kg}\end{array}$$

The percent errors involved in the saturated liquid approximation are listed above in parentheses.

3-64 A piston–cylinder device contains 0.8 kg of steam at 300°C and 1 MPa. Steam is cooled at constant pressure until one-half of the mass condenses.

- (a) Show the process on a T - v diagram.
- (b) Find the final temperature.
- (c) Determine the volume change.

Analysis (b) At the final state the cylinder contains saturated liquid-vapor mixture, and thus the final temperature must be the saturation temperature at the final pressure,

$$T = T_{\text{sat}@1 \text{ MPa}} = \mathbf{179.88^\circ\text{C}} \quad (\text{Table A-5})$$

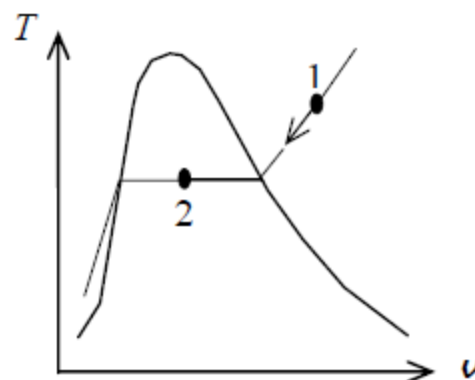
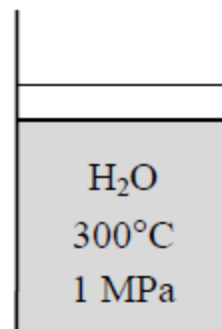
(c) The quality at the final state is specified to be $x_2 = 0.5$. The specific volumes at the initial and the final states are

$$\left. \begin{array}{l} P_1 = 1.0 \text{ MPa} \\ T_1 = 300^\circ\text{C} \end{array} \right\} \nu_1 = 0.25799 \text{ m}^3/\text{kg} \quad (\text{Table A-6})$$

$$\left. \begin{array}{l} P_2 = 1.0 \text{ MPa} \\ x_2 = 0.5 \end{array} \right\} \begin{aligned} \nu_2 &= \nu_f + x_2 \nu_{fg} \\ &= 0.001127 + 0.5 \times (0.19436 - 0.001127) \\ &= 0.09775 \text{ m}^3/\text{kg} \end{aligned}$$

Thus,

$$\Delta V = m(\nu_2 - \nu_1) = (0.8 \text{ kg})(0.09775 - 0.25799) \text{ m}^3/\text{kg} = \mathbf{-0.1282 \text{ m}^3}$$



3-65 A rigid tank contains water vapor at 250°C and an unknown pressure. When the tank is cooled to 150°C, the vapor starts condensing. Estimate the initial pressure in the tank. *Answer: 0.60 MPa*

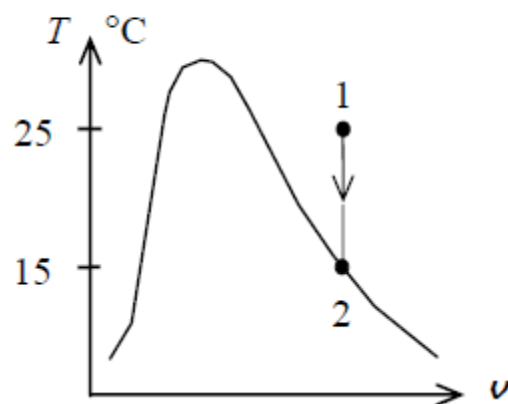
Analysis This is a constant volume process ($v = V/m = \text{constant}$), and the initial specific volume is equal to the final specific volume that is

$$v_1 = v_2 = v_{g@150^\circ\text{C}} = 0.39248 \text{ m}^3/\text{kg} \quad (\text{Table A-4})$$

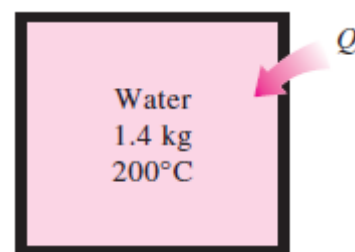
since the vapor starts condensing at 150°C. Then from Table A-6,

$$\left. \begin{array}{l} T_1 = 250^\circ\text{C} \\ v_1 = 0.39248 \text{ m}^3/\text{kg} \end{array} \right\} P_1 = 0.60 \text{ MPa}$$

H_2O $T_1 = 250^\circ\text{C}$ $P_1 = ?$
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3-67 A rigid tank initially contains 1.4-kg saturated liquid water at 200°C. At this state, 25 percent of the volume is occupied by water and the rest by air. Now heat is supplied to the water until the tank contains saturated vapor only. Determine (a) the volume of the tank, (b) the final temperature and pressure, and (c) the internal energy change of the water.



Properties The saturated liquid properties of water at 200°C are: $\nu_f = 0.001157 \text{ m}^3/\text{kg}$ and $u_f = 850.46 \text{ kJ/kg}$ (Table A-4).

Analysis (a) The cylinder initially contains saturated liquid water. The volume of the cylinder at the initial state is

$$V_1 = m\nu_1 = (1.4 \text{ kg})(0.001157 \text{ m}^3/\text{kg}) = 0.001619 \text{ m}^3$$

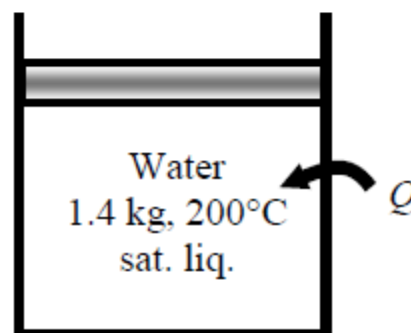
The volume at the final state is

$$V = 4(0.001619) = \mathbf{0.006476 \text{ m}^3}$$

(b) The final state properties are

$$\nu_2 = \frac{V}{m} = \frac{0.006476 \text{ m}^3}{1.4 \text{ kg}} = 0.004626 \text{ m}^3/\text{kg}$$

$$\left. \begin{array}{l} \nu_2 = 0.004626 \text{ m}^3/\text{kg} \\ x_2 = 1 \end{array} \right\} \begin{array}{l} T_2 = \mathbf{371.3^\circ\text{C}} \\ P_2 = \mathbf{21,367 \text{ kPa}} \\ u_2 = 2201.5 \text{ kJ/kg} \end{array} \quad (\text{Table A-4 or A-5 or EES})$$



(c) The total internal energy change is determined from

$$\Delta U = m(u_2 - u_1) = (1.4 \text{ kg})(2201.5 - 850.46) \text{ kJ/kg} = \mathbf{1892 \text{ kJ}}$$

3-68 A piston–cylinder device initially contains steam at 3.5 MPa, superheated by 5°C. Now, steam loses heat to the surroundings and the piston moves down hitting a set of stops at which point the cylinder contains saturated liquid water. The cooling continues until the cylinder contains water at 200°C. Determine (a) the initial temperature, (b) the enthalpy change per unit mass of the steam by the time the piston first hits the stops, and (c) the final pressure and the quality (if mixture).

Analysis (a) The saturation temperature of steam at 3.5 MPa is

$$T_{\text{sat}@3.5 \text{ MPa}} = 242.6^\circ\text{C} \quad (\text{Table A-5})$$

Then, the initial temperature becomes

$$T_1 = 242.6 + 5 = \mathbf{247.6^\circ\text{C}}$$

$$\text{Also, } \left. \begin{array}{l} P_1 = 3.5 \text{ MPa} \\ T_1 = 247.6^\circ\text{C} \end{array} \right\} h_1 = 2821.1 \text{ kJ/kg} \quad (\text{Table A-6})$$

(b) The properties of steam when the piston first hits the stops are

$$\left. \begin{array}{l} P_2 = P_1 = 3.5 \text{ MPa} \\ x_2 = 0 \end{array} \right\} \begin{array}{l} h_2 = 1049.7 \text{ kJ/kg} \\ v_2 = 0.001235 \text{ m}^3/\text{kg} \end{array} \quad (\text{Table A-5})$$

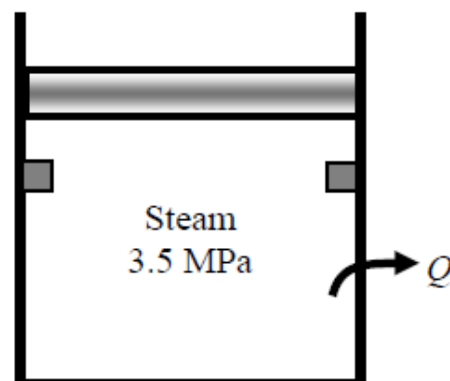
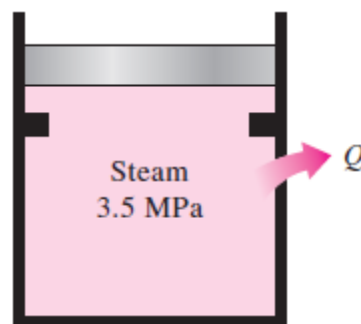
Then, the enthalpy change of steam becomes

$$\Delta h = h_2 - h_1 = 1049.7 - 2821.1 = \mathbf{-1771 \text{ kJ/kg}}$$

(c) At the final state

$$\left. \begin{array}{l} v_3 = v_2 = 0.001235 \text{ m}^3/\text{kg} \\ T_3 = 200^\circ\text{C} \end{array} \right\} \begin{array}{l} P_3 = \mathbf{1555 \text{ kPa}} \\ x_3 = \mathbf{0.0006} \end{array} \quad (\text{Table A-4 or EES})$$

The cylinder contains saturated liquid–vapor mixture with a small mass of vapor at the final state.



3-73 A spherical balloon with a diameter of 6 m is filled with helium at 20°C and 200 kPa. Determine the mole number and the mass of the helium in the balloon. *Answers: 9.28 kmol, 37.15 kg*

Assumptions At specified conditions, helium behaves as an ideal gas.

Properties The universal gas constant is $R_u = 8.314 \text{ kPa}\cdot\text{m}^3/\text{kmol}\cdot\text{K}$. The molar mass of helium is 4.0 kg/kmol (Table A-1).

Analysis The volume of the sphere is

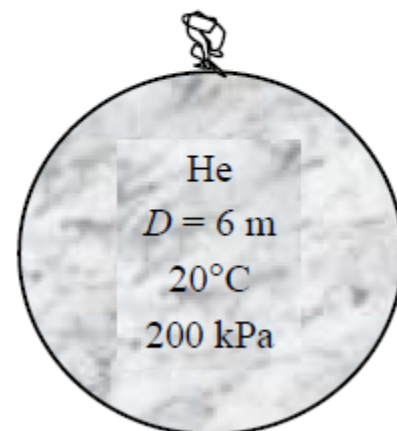
$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (3 \text{ m})^3 = 113.1 \text{ m}^3$$

Assuming ideal gas behavior, the mole numbers of He is determined from

$$N = \frac{PV}{R_u T} = \frac{(200 \text{ kPa})(113.1 \text{ m}^3)}{(8.314 \text{ kPa}\cdot\text{m}^3/\text{kmol}\cdot\text{K})(293 \text{ K})} = \mathbf{9.28 \text{ kmol}}$$

Then the mass of He can be determined from

$$m = NM = (9.28 \text{ kmol})(4.0 \text{ kg/kmol}) = \mathbf{37.15 \text{ kg}}$$



3-75 The pressure in an automobile tire depends on the temperature of the air in the tire. When the air temperature is 25°C, the pressure gage reads 210 kPa. If the volume of the tire is 0.025 m³, determine the pressure rise in the tire when the air temperature in the tire rises to 50°C. Also, determine the amount of air that must be bled off to restore pressure to its original value at this temperature. Assume the atmospheric pressure is 100 kPa.

Assumptions 1 At specified conditions, air behaves as an ideal gas. 2 The volume of the tire remains constant.

Properties The gas constant of air is $R = 0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$ (Table A-1).

Analysis Initially, the absolute pressure in the tire is

$$P_1 = P_g + P_{\text{atm}} = 210 + 100 = 310 \text{ kPa}$$

Treating air as an ideal gas and assuming the volume of the tire to remain constant, the final pressure in the tire can be determined from

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \longrightarrow P_2 = \frac{T_2}{T_1} P_1 = \frac{323 \text{ K}}{298 \text{ K}} (310 \text{ kPa}) = 336 \text{ kPa}$$

Thus the pressure rise is

$$\Delta P = P_2 - P_1 = 336 - 310 = \mathbf{26 \text{ kPa}}$$

The amount of air that needs to be bled off to restore pressure to its original value is

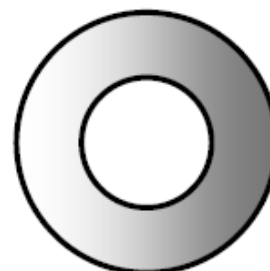
$$m_1 = \frac{P_1 V}{RT_1} = \frac{(310 \text{ kPa})(0.025 \text{ m}^3)}{(0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(298 \text{ K})} = 0.0906 \text{ kg}$$

$$m_2 = \frac{P_1 V}{RT_2} = \frac{(310 \text{ kPa})(0.025 \text{ m}^3)}{(0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(323 \text{ K})} = 0.0836 \text{ kg}$$

$$\Delta m = m_1 - m_2 = 0.0906 - 0.0836 = \mathbf{0.0070 \text{ kg}}$$

Tire

25°C



3–80 A 1-m³ tank containing air at 25°C and 500 kPa is connected through a valve to another tank containing 5 kg of air at 35°C and 200 kPa. Now the valve is opened, and the entire system is allowed to reach thermal equilibrium with the surroundings, which are at 20°C. Determine the volume of the second tank and the final equilibrium pressure of air.

Answers: 2.21 m³, 284.1 kPa

Assumptions At specified conditions, air behaves as an ideal gas.

Properties The gas constant of air is $R = 0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$ (Table A-1).

Analysis Let's call the first and the second tanks A and B. Treating air as an ideal gas, the volume of the second tank and the mass of air in the first tank are determined to be

$$V_B = \left(\frac{m_1 R T_1}{P_1} \right)_B = \frac{(5 \text{ kg})(0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(308 \text{ K})}{200 \text{ kPa}} = 2.21 \text{ m}^3$$

$$m_A = \left(\frac{P_1 V}{R T_1} \right)_A = \frac{(500 \text{ kPa})(1.0 \text{ m}^3)}{(0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(298 \text{ K})} = 5.846 \text{ kg}$$

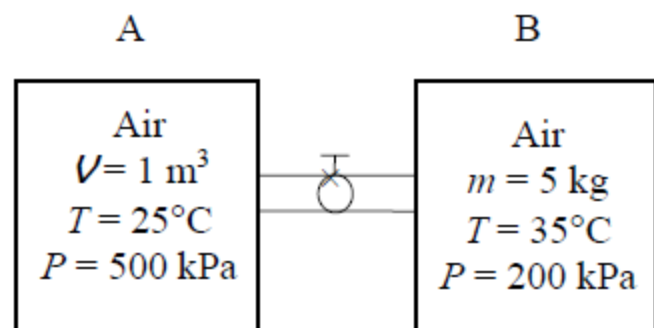
Thus,

$$V = V_A + V_B = 1.0 + 2.21 = 3.21 \text{ m}^3$$

$$m = m_A + m_B = 5.846 + 5.0 = 10.846 \text{ kg}$$

Then the final equilibrium pressure becomes

$$P_2 = \frac{m R T_2}{V} = \frac{(10.846 \text{ kg})(0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(293 \text{ K})}{3.21 \text{ m}^3} = 284.1 \text{ kPa}$$



3-84 Determine the specific volume of superheated water vapor at 10 MPa and 400°C, using (a) the ideal-gas equation, (b) the generalized compressibility chart, and (c) the steam tables. Also determine the error involved in the first two cases.

Answers: (a) 0.03106 m³/kg, 17.6 percent; (b) 0.02609 m³/kg, 1.2 percent; (c) 0.02644 m³/kg

Properties The gas constant, the critical pressure, and the critical temperature of water are, from Table A-1,

$$R = 0.4615 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}, \quad T_{cr} = 647.1 \text{ K}, \quad P_{cr} = 22.06 \text{ MPa}$$

Analysis (a) From the ideal gas equation of state,

$$\nu = \frac{RT}{P} = \frac{(0.4615 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(673 \text{ K})}{(10,000 \text{ kPa})} = \mathbf{0.03106 \text{ m}^3/\text{kg} \text{ (17.6\% error)}}$$

(b) From the compressibility chart (Fig. A-15),

$$\left. \begin{aligned} P_R &= \frac{P}{P_{cr}} = \frac{10 \text{ MPa}}{22.06 \text{ MPa}} = 0.453 \\ T_R &= \frac{T}{T_{cr}} = \frac{673 \text{ K}}{647.1 \text{ K}} = 1.04 \end{aligned} \right\} Z = 0.84$$

H₂O 10 MPa 400°C
--

Thus,

$$\nu = Z\nu_{ideal} = (0.84)(0.03106 \text{ m}^3/\text{kg}) = \mathbf{0.02609 \text{ m}^3/\text{kg} \text{ (1.2\% error)}}$$

(c) From the superheated steam table (Table A-6),

$$\left. \begin{aligned} P &= 10 \text{ MPa} \\ T &= 400^\circ\text{C} \end{aligned} \right\} \nu = \mathbf{0.02644 \text{ m}^3/\text{kg}}$$

3-86 Determine the specific volume of refrigerant-134a vapor at 0.9 MPa and 70°C based on (a) the ideal-gas equation, (b) the generalized compressibility chart, and (c) data from tables. Also, determine the error involved in the first two cases.

Properties The gas constant, the critical pressure, and the critical temperature of refrigerant-134a are, from Table A-1,

$$R = 0.08149 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}, \quad T_{cr} = 374.2 \text{ K}, \quad P_{cr} = 4.059 \text{ MPa}$$

Analysis (a) From the ideal gas equation of state,

$$\nu = \frac{RT}{P} = \frac{(0.08149 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(343 \text{ K})}{900 \text{ kPa}} = \mathbf{0.03105 \text{ m}^3/\text{kg}} \quad (13.3\% \text{ error})$$

(b) From the compressibility chart (Fig. A-15),

$$\left. \begin{aligned} P_R &= \frac{P}{P_{cr}} = \frac{0.9 \text{ MPa}}{4.059 \text{ MPa}} = 0.222 \\ T_R &= \frac{T}{T_{cr}} = \frac{343 \text{ K}}{374.2 \text{ K}} = 0.917 \end{aligned} \right\} Z = 0.894$$

R-134a
0.9 MPa
70°C

Thus,

$$\nu = Z\nu_{ideal} = (0.894)(0.03105 \text{ m}^3/\text{kg}) = \mathbf{0.02776 \text{ m}^3/\text{kg}} \quad (1.3\% \text{ error})$$

(c) From the superheated refrigerant table (Table A-13),

$$\left. \begin{aligned} P &= 0.9 \text{ MPa} \\ T &= 70^\circ\text{C} \end{aligned} \right\} \nu = \mathbf{0.027413 \text{ m}^3/\text{kg}}$$

3-87 Determine the specific volume of nitrogen gas at 10 MPa and 150 K based on (a) the ideal-gas equation and (b) the generalized compressibility chart. Compare these results with the experimental value of $0.002388 \text{ m}^3/\text{kg}$, and determine the error involved in each case. *Answers: (a) $0.004452 \text{ m}^3/\text{kg}$, 86.4 percent; (b) $0.002404 \text{ m}^3/\text{kg}$, 0.7 percent*

Properties The gas constant, the critical pressure, and the critical temperature of nitrogen are, from Table A-1,

$$R = 0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}, \quad T_{\text{cr}} = 126.2 \text{ K}, \quad P_{\text{cr}} = 3.39 \text{ MPa}$$

Analysis (a) From the ideal gas equation of state,

$$\nu = \frac{RT}{P} = \frac{(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(150 \text{ K})}{10,000 \text{ kPa}} = 0.004452 \text{ m}^3/\text{kg} \quad (86.4\% \text{ error})$$

(b) From the compressibility chart (Fig. A-15),

$$\left. \begin{aligned} P_R &= \frac{P}{P_{\text{cr}}} = \frac{10 \text{ MPa}}{3.39 \text{ MPa}} = 2.95 \\ T_R &= \frac{T}{T_{\text{cr}}} = \frac{150 \text{ K}}{126.2 \text{ K}} = 1.19 \end{aligned} \right\} Z = 0.54$$

N₂
10 MPa
150 K

Thus,

$$\nu = Z\nu_{\text{ideal}} = (0.54)(0.004452 \text{ m}^3/\text{kg}) = 0.002404 \text{ m}^3/\text{kg} \quad (0.7\% \text{ error})$$

3–88 Determine the specific volume of superheated water vapor at 3.5 MPa and 450°C based on (a) the ideal-gas equation, (b) the generalized compressibility chart, and (c) the steam tables. Determine the error involved in the first two cases.

Properties The gas constant, the critical pressure, and the critical temperature of water are, from Table A-1,

$$R = 0.4615 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}, \quad T_{cr} = 647.1 \text{ K}, \quad P_{cr} = 22.06 \text{ MPa}$$

Analysis (a) From the ideal gas equation of state,

$$\nu = \frac{RT}{P} = \frac{(0.4615 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(723 \text{ K})}{3500 \text{ kPa}} = \mathbf{0.09533 \text{ m}^3/\text{kg}} \quad (3.7\% \text{ error})$$

(b) From the compressibility chart (Fig. A-15),

$$\left. \begin{aligned} P_R &= \frac{P}{P_{cr}} = \frac{3.5 \text{ MPa}}{22.06 \text{ MPa}} = 0.159 \\ T_R &= \frac{T}{T_{cr}} = \frac{723 \text{ K}}{647.1 \text{ K}} = 1.12 \end{aligned} \right\} Z = 0.961$$

H₂O
3.5 MPa
450°C

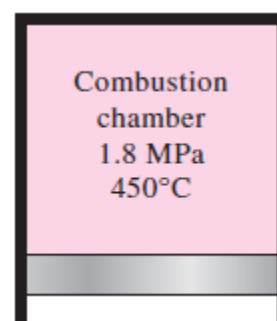
Thus,

$$\nu = Z\nu_{ideal} = (0.961)(0.09533 \text{ m}^3/\text{kg}) = \mathbf{0.09161 \text{ m}^3/\text{kg}} \quad (0.4\% \text{ error})$$

(c) From the superheated steam table (Table A-6),

$$\left. \begin{aligned} P &= 3.5 \text{ MPa} \\ T &= 450^\circ\text{C} \end{aligned} \right\} \nu = \mathbf{0.09196 \text{ m}^3/\text{kg}}$$

3–108 The combustion in a gasoline engine may be approximated by a constant volume heat addition process. There exists the air–fuel mixture in the cylinder before the combustion and the combustion gases after it, and both may be approximated as air, an ideal gas. In a gasoline engine, the cylinder conditions are 1.8 MPa and 450°C before the combustion and 1300°C after it. Determine the pressure at the end of the combustion process. *Answer: 3916 kPa*



Assumptions 1 The contents of cylinder are approximated by the air properties. 2 Air is an ideal gas.

Analysis The final pressure may be determined from the ideal gas relation

$$P_2 = \frac{T_2}{T_1} P_1 = \left(\frac{1300 + 273 \text{ K}}{450 + 273 \text{ K}} \right) (1800 \text{ kPa}) = \mathbf{3916 \text{ kPa}}$$

3-109 A rigid tank contains an ideal gas at 300 kPa and 600 K. Now half of the gas is withdrawn from the tank and the gas is found at 100 kPa at the end of the process. Determine (a) the final temperature of the gas and (b) the final pressure if no mass was withdrawn from the tank and the same final temperature was reached at the end of the process.

Analysis (a) The first case is a constant volume process. When half of the gas is withdrawn from the tank, the final temperature may be determined from the ideal gas relation as

$$T_2 = \frac{m_1}{m_2} \frac{P_2}{P_1} T_1 = (2) \left(\frac{100 \text{ kPa}}{300 \text{ kPa}} \right) (600 \text{ K}) = \mathbf{400 \text{ K}}$$

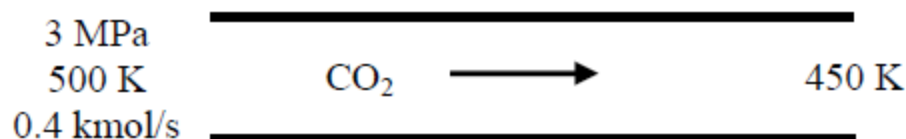
(b) The second case is a constant volume and constant mass process. The ideal gas relation for this case yields

$$P_2 = \frac{T_2}{T_1} P_1 = \left(\frac{400 \text{ K}}{600 \text{ K}} \right) (300 \text{ kPa}) = \mathbf{200 \text{ kPa}}$$

Ideal gas 300 kPa 600 K

3-110 Carbon-dioxide gas at 3 MPa and 500 K flows steadily in a pipe at a rate of 0.4 kmol/s. Determine (a) the volume and mass flow rates and the density of carbon dioxide at this state. If CO₂ is cooled at constant pressure as it flows in the pipe so that the temperature of CO₂ drops to 450 K at the exit of the pipe, determine (b) the volume flow rate at the exit of the pipe.

Analysis (a) The volume and mass flow rates may be determined from ideal gas relation as



$$\dot{V}_1 = \frac{\dot{N}R_u T_1}{P} = \frac{(0.4 \text{ kmol/s})(8.314 \text{ kPa}\cdot\text{m}^3/\text{kmol}\cdot\text{K})(500 \text{ K})}{3000 \text{ kPa}} = \mathbf{0.5543 \text{ m}^3/\text{s}}$$

$$\dot{m}_1 = \frac{P_1 \dot{V}_1}{RT_1} = \frac{(3000 \text{ kPa})(0.5543 \text{ m}^3/\text{s})}{(0.1889 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(500 \text{ K})} = \mathbf{17.60 \text{ kg/s}}$$

The density is

$$\rho_1 = \frac{\dot{m}_1}{\dot{V}_1} = \frac{(17.60 \text{ kg/s})}{(0.5543 \text{ m}^3/\text{s})} = \mathbf{31.76 \text{ kg/m}^3}$$

(b) The volume flow rate at the exit is

$$\dot{V}_2 = \frac{\dot{N}R_u T_2}{P} = \frac{(0.4 \text{ kmol/s})(8.314 \text{ kPa}\cdot\text{m}^3/\text{kmol}\cdot\text{K})(450 \text{ K})}{3000 \text{ kPa}} = \mathbf{0.4988 \text{ m}^3/\text{s}}$$

3-112 Combustion in a diesel engine may be modeled as a constant-pressure heat addition process with air in the cylinder before and after combustion. Consider a diesel engine with cylinder conditions of 950 K and 75 cm³ before combustion, and 150 cm³ after it. The engine operates with an air–fuel ratio of 22 kg air/kg fuel (the mass of the air divided by the mass of the fuel). Determine the temperature after the combustion process.

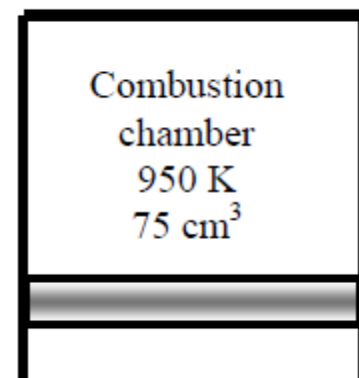
Assumptions 1 The contents of cylinder is approximated by the air properties. 2 Air is an ideal gas.

Analysis The ratio of the initial to the final mass is

$$\frac{m_1}{m_2} = \frac{AF}{AF + 1} = \frac{22}{22 + 1} = \frac{22}{23}$$

The final temperature may be determined from ideal gas relation

$$T_2 = \frac{m_1}{m_2} \frac{V_2}{V_1} T_1 = \left(\frac{22}{23} \right) \left(\frac{150 \text{ cm}^3}{75 \text{ cm}^3} \right) (950 \text{ K}) = \mathbf{1817 \text{ K}}$$



3-120 A 4-L rigid tank contains 2 kg of saturated liquid–vapor mixture of water at 50°C. The water is now slowly heated until it exists in a single phase. At the final state, will the water be in the liquid phase or the vapor phase? What would your answer be if the volume of the tank were 400 L instead of 4 L?

Analysis This is a constant volume process ($\nu = V/m = \text{constant}$), and thus the final specific volume will be equal to the initial specific volume,

$$\nu_2 = \nu_1$$

The critical specific volume of water is 0.003106 m³/kg. Thus if the final specific volume is smaller than this value, the water will exist as a liquid, otherwise as a vapor.

$$V = 4\text{ L} \longrightarrow \nu = \frac{V}{m} = \frac{0.004\text{ m}^3}{2\text{ kg}} = 0.002\text{ m}^3/\text{kg} < \nu_{\text{cr}} \quad \text{Thus, liquid.}$$

$$V = 400\text{ L} \longrightarrow \nu = \frac{V}{m} = \frac{0.4\text{ m}^3}{2\text{ kg}} = 0.2\text{ m}^3/\text{kg} > \nu_{\text{cr}}. \quad \text{Thus, vapor.}$$

H ₂ O
$V = 4\text{ L}$
$m = 2\text{ kg}$
$T = 50^\circ\text{C}$

3-122 A 0.5-m³ rigid tank containing hydrogen at 20°C and 600 kPa is connected by a valve to another 0.5-m³ rigid tank that holds hydrogen at 30°C and 150 kPa. Now the valve is opened and the system is allowed to reach thermal equilibrium with the surroundings, which are at 15°C. Determine the final pressure in the tank.

Properties The gas constant for hydrogen is 4.124 kPa·m³/kg·K (Table A-1).

Analysis Let's call the first and the second tanks A and B. Treating H₂ as an ideal gas, the total volume and the total mass of H₂ are

$$V = V_A + V_B = 0.5 + 0.5 = 1.0 \text{ m}^3$$

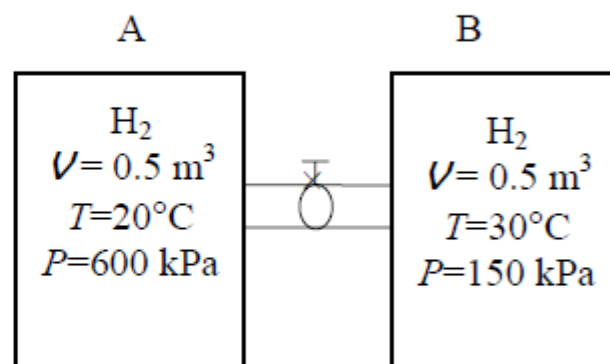
$$m_A = \left(\frac{P_1 V}{RT_1} \right)_A = \frac{(600 \text{ kPa})(0.5 \text{ m}^3)}{(4.124 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(293 \text{ K})} = 0.248 \text{ kg}$$

$$m_B = \left(\frac{P_1 V}{RT_1} \right)_B = \frac{(150 \text{ kPa})(0.5 \text{ m}^3)}{(4.124 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(303 \text{ K})} = 0.060 \text{ kg}$$

$$m = m_A + m_B = 0.248 + 0.060 = 0.308 \text{ kg}$$

Then the final pressure can be determined from

$$P = \frac{mRT_2}{V} = \frac{(0.308 \text{ kg})(4.124 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(288 \text{ K})}{1.0 \text{ m}^3} = 365.8 \text{ kPa}$$



3-124 A 20-m³ tank contains nitrogen at 23°C and 600 kPa. Some nitrogen is allowed to escape until the pressure in the tank drops to 400 kPa. If the temperature at this point is 20°C, determine the amount of nitrogen that has escaped.

Answer: 44.6 kg

Properties The gas constant for nitrogen is 0.2968 kPa·m³/kg·K (Table A-1).

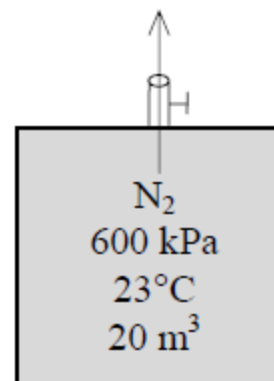
Analysis Treating N₂ as an ideal gas, the initial and the final masses in the tank are determined to be

$$m_1 = \frac{P_1 V}{RT_1} = \frac{(600 \text{ kPa})(20 \text{ m}^3)}{(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(296 \text{ K})} = 136.6 \text{ kg}$$

$$m_2 = \frac{P_2 V}{RT_2} = \frac{(400 \text{ kPa})(20 \text{ m}^3)}{(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(293 \text{ K})} = 92.0 \text{ kg}$$

Thus the amount of N₂ that escaped is

$$\Delta m = m_1 - m_2 = 136.6 - 92.0 = \mathbf{44.6 \text{ kg}}$$



3-125 Steam at 400°C has a specific volume of 0.02 m³/kg. Determine the pressure of the steam based on (a) the ideal-gas equation, (b) the generalized compressibility chart, and (c) the steam tables. *Answers: (a) 15,529 kPa, (b) 12,576 kPa, (c) 12,500 kPa*

Properties The gas constant, the critical pressure, and the critical temperature of water are, from Table A-1,

$$R = 0.4615 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}, \quad T_{\text{cr}} = 647.1 \text{ K}, \quad P_{\text{cr}} = 22.06 \text{ MPa}$$

Analysis (a) From the ideal gas equation of state,

$$P = \frac{RT}{\nu} = \frac{(0.4615 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(673 \text{ K})}{0.02 \text{ m}^3/\text{kg}} = \mathbf{15,529 \text{ kPa}}$$

(b) From the compressibility chart (Fig. A-15a),

$$\left. \begin{aligned} T_R &= \frac{T}{T_{\text{cr}}} = \frac{673 \text{ K}}{647.1 \text{ K}} = 1.040 \\ \nu_R &= \frac{\nu_{\text{actual}}}{RT_{\text{cr}}/P_{\text{cr}}} = \frac{(0.02 \text{ m}^3/\text{kg})(22,060 \text{ kPa})}{(0.4615 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(647.1 \text{ K})} = 1.48 \end{aligned} \right\} P_R = 0.57$$

Thus, $P = P_R P_{\text{cr}} = 0.57 \times 22,060 = \mathbf{12,574 \text{ kPa}}$

(c) From the superheated steam table,

$$\left. \begin{aligned} T &= 400^\circ\text{C} \\ \nu &= 0.02 \text{ m}^3/\text{kg} \end{aligned} \right\} P = \mathbf{12,576 \text{ kPa}} \quad (\text{from EES})$$

H_2O $0.02 \text{ m}^3/\text{kg}$ 400°C

3-126 A tank whose volume is unknown is divided into two parts by a partition. One side of the tank contains 0.01 m^3 of refrigerant-134a that is a saturated liquid at 0.8 MPa , while the other side is evacuated. The partition is now removed, and the refrigerant fills the entire tank. If the final state of the refrigerant is 20°C and 400 kPa , determine the volume of the tank.

Analysis The mass of the refrigerant contained in the tank is

$$m = \frac{V_1}{v_1} = \frac{0.01 \text{ m}^3}{0.0008458 \text{ m}^3/\text{kg}} = 11.82 \text{ kg}$$

since

$$v_1 = v_{f@0.8\text{MPa}} = 0.0008458 \text{ m}^3/\text{kg}$$

At the final state (Table A-13),

$$\left. \begin{array}{l} P_2 = 400 \text{ kPa} \\ T_2 = 20^\circ\text{C} \end{array} \right\} v_2 = 0.05421 \text{ m}^3/\text{kg}$$

Thus, $V_{\text{tank}} = V_2 = m v_2 = (11.82 \text{ kg})(0.05421 \text{ m}^3/\text{kg}) = \mathbf{0.641 \text{ m}^3}$

R-134a $P=0.8 \text{ MPa}$ $V=0.01 \text{ m}^3$	Evacuated
---	-----------

3-128 Liquid propane is commonly used as a fuel for heating homes, powering vehicles such as forklifts, and filling portable picnic tanks. Consider a propane tank that initially contains 5 L of liquid propane at the environment temperature of 20°C. If a hole develops in the connecting tube of a propane tank and the propane starts to leak out, determine the temperature of propane when the pressure in the tank drops to 1 atm. Also, determine the total amount of heat transfer from the environment to the tank to vaporize the entire propane in the tank.

Properties The properties of propane at 1 atm are $T_{\text{sat}} = -42.1^\circ\text{C}$, $\rho = 581 \text{ kg/m}^3$, and $h_{\text{fg}} = 427.8 \text{ kJ/kg}$ (Table A-3).

Analysis The temperature of propane when the pressure drops to 1 atm is simply the saturation pressure at that temperature,

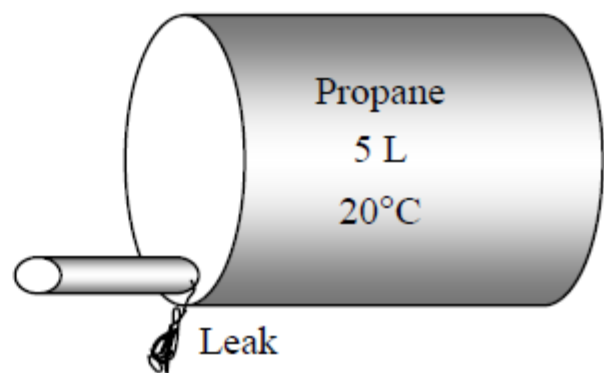
$$T = T_{\text{sat}@1 \text{ atm}} = -42.1^\circ\text{C}$$

The initial mass of liquid propane is

$$m = \rho V = (581 \text{ kg/m}^3)(0.005 \text{ m}^3) = 2.905 \text{ kg}$$

The amount of heat absorbed is simply the total heat of vaporization,

$$Q_{\text{absorbed}} = mh_{\text{fg}} = (2.905 \text{ kg})(427.8 \text{ kJ/kg}) = 1243 \text{ kJ}$$



3-129 Repeat Prob. 3-128 for isobutane.

Properties The properties of isobutane at 1 atm are $T_{\text{sat}} = -11.7^\circ\text{C}$, $\rho = 593.8 \text{ kg/m}^3$, and $h_{\text{fg}} = 367.1 \text{ kJ/kg}$ (Table A-3).

Analysis The temperature of isobutane when the pressure drops to 1 atm is simply the saturation pressure at that temperature,

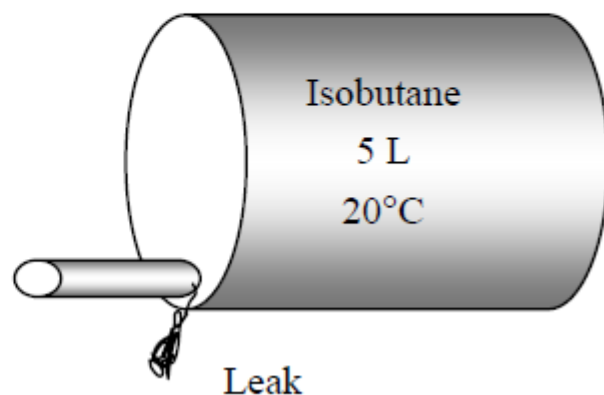
$$T = T_{\text{sat}@1 \text{ atm}} = \mathbf{-11.7^\circ\text{C}}$$

The initial mass of liquid isobutane is

$$m = \rho V = (593.8 \text{ kg/m}^3)(0.005 \text{ m}^3) = 2.969 \text{ kg}$$

The amount of heat absorbed is simply the total heat of vaporization,

$$Q_{\text{absorbed}} = mh_{\text{fg}} = (2.969 \text{ kg})(367.1 \text{ kJ/kg}) = \mathbf{1090 \text{ kJ}}$$



3-130 A tank contains helium at 100°C and 10 kPa gage. The helium is heated in a process by heat transfer from the surroundings such that the helium reaches a final equilibrium state at 300°C. Determine the final gage pressure of the helium. Assume atmospheric pressure is 100 kPa.

Assumptions 1 Helium is an ideal gas.

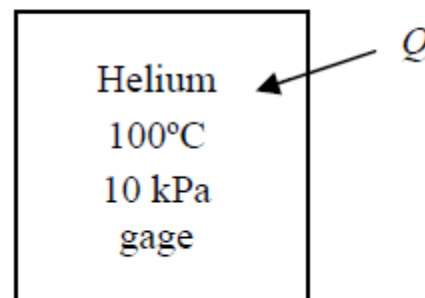
Properties The local atmospheric pressure is given to be 100 kPa.

Analysis Noting that the specific volume of helium in the tank remains constant, from ideal gas relation, we have

$$P_2 = P_1 \frac{T_2}{T_1} = (10 + 100 \text{ kPa}) \frac{(300 + 273)\text{K}}{(100 + 273)\text{K}} = 169.0 \text{ kPa}$$

Then the gage pressure becomes

$$P_{\text{gage},2} = P_2 - P_{\text{atm}} = 169.0 - 100 = \mathbf{69.0 \text{ kPa}}$$



3-132 Complete the blank cells in the following table of properties of steam. In the last column describe the condition of steam as compressed liquid, saturated mixture, superheated vapor, or insufficient information; and, if applicable, give the quality.

P , kPa	T , °C	ν , m ³ /kg	u , kJ/kg	Condition description and quality (if applicable)
200	30			
270.3	130			
	400	1.5493		
300		0.500		
500			3084	

P , kPa	T , °C	ν , m ³ /kg	u , kJ/kg	Phase description
200	30	0.001004	125.71	Compressed liquid
270.3	130	-	-	Insufficient information
200	400	1.5493	2967.2	Superheated steam
300	133.52	0.500	2196.4	Saturated mixture, x=0.825
500	473.1	0.6858	3084	Superheated steam

3-133 Complete the blank cells in the following table of properties of refrigerant-134a. In the last column describe the condition of refrigerant-134a as compressed liquid, saturated mixture, superheated vapor, or insufficient information; and, if applicable, give the quality.

P , kPa	T , °C	ν , m ³ /kg	u , kJ/kg	Condition description and quality (if applicable)
320	-12			
1000	39.37			
	40	0.17794		
180		0.0700		
200			249	

P , kPa	T , °C	ν , m ³ /kg	u , kJ/kg	Phase description
320	-12	0.0007497	35.72	Compressed liquid
1000	39.37	-	-	Insufficient information
140	40	0.17794	263.79	Superheated vapor
180	-12.73	0.0700	153.66	Saturated mixture, x=0.6315
200	22.13	0.1152	249	Superheated vapor

Chapter 3

End of the Problems